



ROBOCUP RESCUE LINE JUNIOR TEAM DESCRIPTION PAPER

Antares



General information for this template

- This template contains the structure for your Team Description Paper. Please look into the official rubrics posted on the official RoboCup website and the community website to see which areas of your TDP will be scored.

- The document should be a maximum of 10 pages long (from Abstract to Conclusion). Please keep the formatting (font size and type, margins, line spacing, etc.), number figures and tables from this template. All text in blue is included for clarity and can be removed. Text in black must remain. It's not necessary to maintain the bullet format. You may write continuous paragraphs within sections.

- Use diagrams, flow charts, etc. throughout this document to better illustrate your work.

- Submit the TDP as a PDF file.

- The competition organizer might ask the team to submit the TDP in a web form. The details for this format would be shared by the competition organizer.

Abstract

- Abstract should be between 150-250 words. Describe your robot, its main capabilities and what sets it apart from competitors. Do not describe your RoboCupJunior Rescue sub-league in detail.

This is our first time participating in RoboCup, so we have no prior experience in this competition. Regarding the robot, we decided to use a QTR-8A (Arduino light sensor) connected to an Arduino Nano, communicating with an EV3. We also used Lego color sensors for green detection, which we were also going to use for line following, but we realized that the color change to infrared could freeze the program, so we opted to change the method for line following.

We learned a lot through trial and error, constantly changing our ideas until we felt it could handle the challenge.

Now, with this sensor, it receives more information about the line, which significantly improves its tracking. It's much more precise and offers many advantages. It's more complex to prevent the robot from getting stuck, and it handles different challenges on the line better. We used four motors, which operate synchronously, with the rear motors following their respective front motors. The power supply was changed to an EV3 battery that provides a constant 9V. Inclined and declined planes are much easier to navigate with all four motors. We had this problem at the beginning of the work for this competition and we worked to solve it, coming to the conclusion that we needed four motors.

1. Introduction

a. Team

Camilo González: I'm sixteen years old and from Canelones, Uruguay. I started competing in robotics in 2025 with a national SUMO tournament.

This is my first time participating in RoboCup. I'm the captain of Team Antares, and my role focuses more on the software, claw design, and robot programming, but we both try to complement each other and do whatever is necessary.

Ismael Bimonte: I'm sixteen years old and from Canelones, Uruguay. I started competing in robotics in 2025 with a national SUMO tournament. This is my first time participating in a RoboCup. My main responsibilities are focused on the robot's hardware, as well as programming and assembling its different parts.

2. Project Planning

a. Overall Project Plan

- Describe your team's objective for the competition 2 ○ Define requirements that your robot and team must meet based on constraints (i.e. physical constraints, time constraints, competition rules, etc.).
- Describe the overall project plan ○ Highlight key milestones with a project schedule/timeline (i.e. include deadlines) ○ Description of each milestone ○ Associate each milestone with a function/team member ○ Include gates to review project progress (if needed)
- Explain how the team agreed on the schedule/plan, examples include: ○ Include analysis of task and competition constraints ○ What is the sequence of milestones and why (i.e. building mechanical robot then programming software) ○ What

conditions will be tested during each iteration ○ Impact of past performance on subsequent iterations/runs

Our team's objective is to design and build a robot that meets the competition's expectations, combining prior knowledge with new learning to achieve satisfactory performance. Furthermore, we aim to develop our ability to solve problems collaboratively, analyzing different proposals until we arrive at solutions that all members agree on. During the project's development, we started with a basic idea that evolved through a continuous process of testing and improvement. This allowed us to build a more reliable, efficient robot capable of facing the challenges posed by the competition. Team members were assigned specific responsibilities, including programming, robot construction, electronics integration, testing, and documentation, ensuring efficient project development and collaboration.

Key project milestones

date	milestone	Responsible
30-March-2026	Rules analysis and project planning	the whole team
4-April-2026	Robot mechanical design	the whole team
13-April-2026	Chassis construction and component assembly	the whole team
3-May-2026	Installation and testing of sensors	the whole team
28-May-2026	Line tracking program development	the whole team
5-June-2026	Hardware and software integration	the whole team
14-June-2026	Track testing and error correction	the whole team
11-June-2026	Final optimization and preparation for competition	the whole team

Each milestone was reviewed before proceeding to the next stage to ensure the project's proper progress.

Requirements Definition

The project requirements stipulated that both devices were compact, durable,

and easy to repair during the competition. It also had to be of an approval size to allow it to move easily around the track and see come the various challenges.

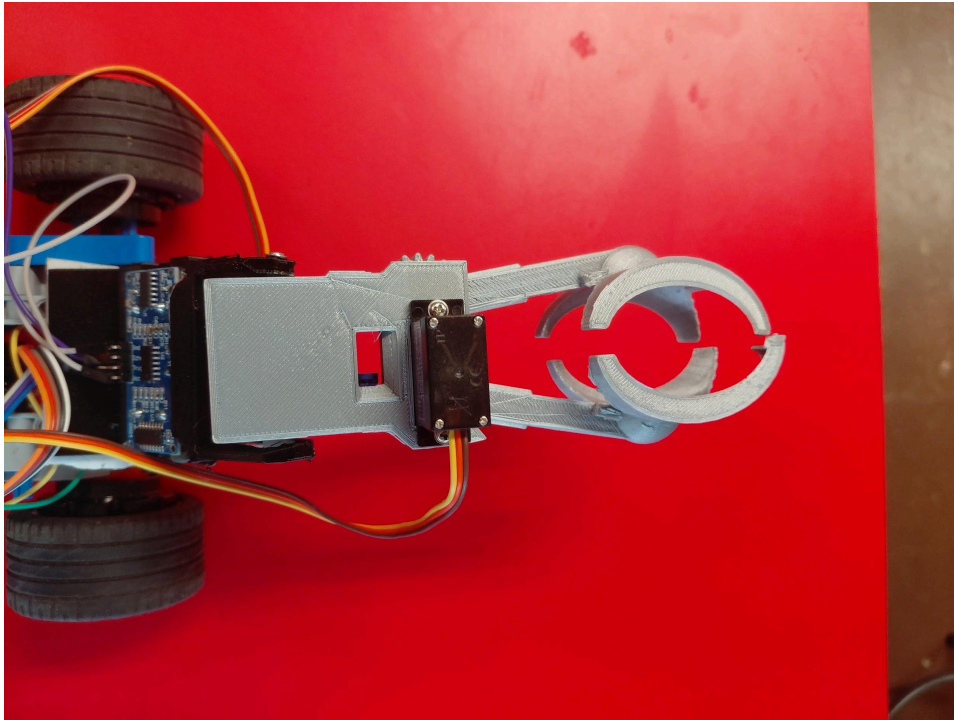
Development time was also taken into account. The goal was to build a fast and efficient robot, capable of meeting all the challenges without speed affecting its performance reliability.

The competition rules served as a guide for the project's organization. The robot had to operate completely autonomously, recognize intersections, identify and classify targets, and, most importantly, follow the line and efficiently.

Integration Plan

To correctly position the motors, several tests and adjustments were necessary to find a configuration that maintained a compact, stable, and efficient structure. Subsequently, the sensors, including the line-following assembly, were installed on the same base, optimizing the available space to incorporate the EV3 Brick, the Arduino Nano, and the gripping mechanism.

The LEGO components communicate through the EV3 Brick, while the EV3 connects to the Arduino Nano via USB serial communication. This integration allowed combining the capabilities of both platforms, expanding the number of sensors and devices that could be used and avoiding the limitations of a single controller. During the integration process, each component was individually tested before being incorporated into the complete system. Once the robot was assembled, functional tests were performed to verify communication between the devices, sensor readings, and the correct operation of the movement and rescue mechanisms.



3-Hardware

- Give a high-level overview of the hardware design of your robot
- Highlight important features and talk about how everything works together

Our robot uses a hybrid architecture that combines the LEGO Mindstorms EV3 platform with an Arduino Nano board. This configuration allows us to leverage the robustness and reliability of the EV3 system along with the flexibility that Arduino offers for integrating custom sensors and electronic components.

The EV3 acts as the main control unit, while the Arduino Nano processes the information from the infrared sensor array installed on the front of the robot. Both devices work in coordination to enable precise navigation and a rapid response to the various situations that may arise during competition.

The robot features a structure based on LEGO Technic pieces complemented by 3D-printed components. It also has a four-wheel configuration to improve stability and weight distribution, as well as a front gripper actuated by a servo motor.

Overall, the hardware was designed to be robust, modular, and adaptable, facilitating both maintenance and future upgrades.

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Mechanical Design and Manufacturing

- Provide a detailed overview of each aspect of the robot's mechanical design. Notably:
 - Main Structure
 - Actuators and Power Train
 - Subassemblies/modules, etc.
 - Rescue mechanism (Line only)
- Provide drawings, diagrams, or images to support your explanations and reasoning for your design choices. Reference requirements outlined during the project planning
 - Explain the testing procedures used to validate the design. Include relevant testing data.
 - Highlight innovative and unusual solutions/approaches. Link back to constraints and requirements.

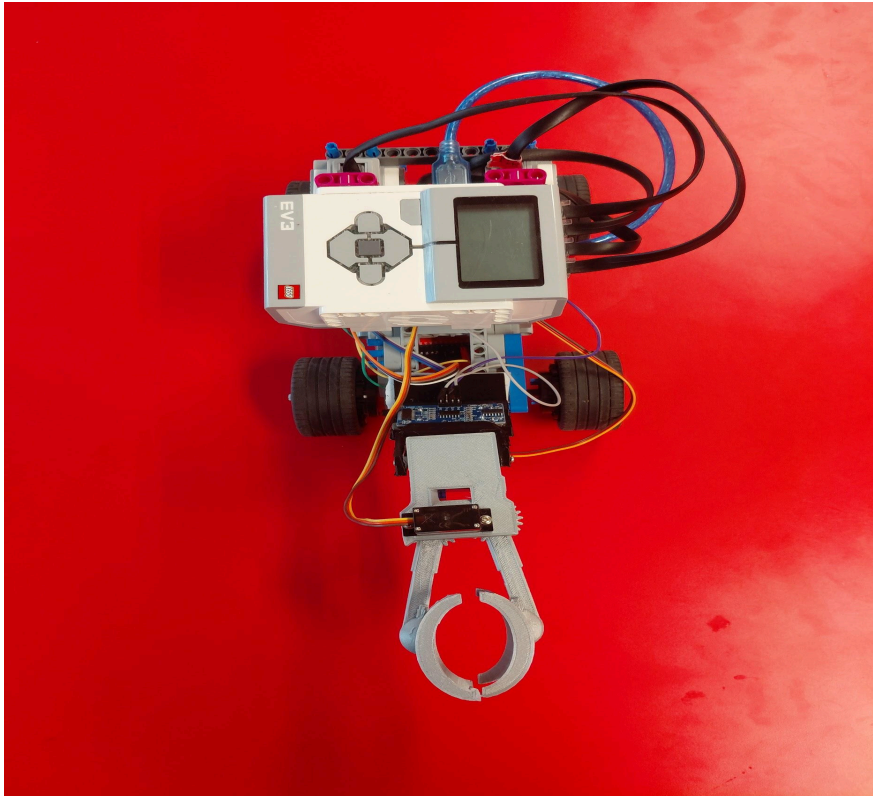
a.

The main structure was built using LEGO Technic pieces, selected for their strength, modularity, and ease of assembly. This approach allowed for rapid modifications during the robot's design, testing, and optimization phases.

The locomotion system consists of four wheels, providing stability and balanced weight distribution. This configuration improves movement precision and facilitates navigation through the various challenges of the course.

In addition, the robot incorporates a 3D-printed rescue mechanism on the front. The mechanism comprises two symmetrical arms powered by an MG90S servo motor and a gear system that transmits movement, enabling precise capture of victims. The use of 3D-printed parts resulted in a lightweight, strong, and easily customizable design.

The combination of LEGO elements and 3D-printed components allowed for the development of a robust mechanical structure adaptable to future modifications. Our robot evolved over the months. The initial concept involved a robot with two motors and a caster wheel. Later, we had to add two more wheels because the robot couldn't climb the inclined plates.



We believe our rescue mechanism is an efficient solution because it was designed prioritizing simplicity and reliability. Our goal was to develop a system capable of consistently performing rescue tasks without incorporating unnecessarily complex mechanisms.

We believe a simpler design reduces the likelihood of failures during competition and facilitates both maintenance and adjustments made during testing. Based on our experience, this approach allows us to obtain a more robust system, better prepared to handle the various situations that may arise on the track.

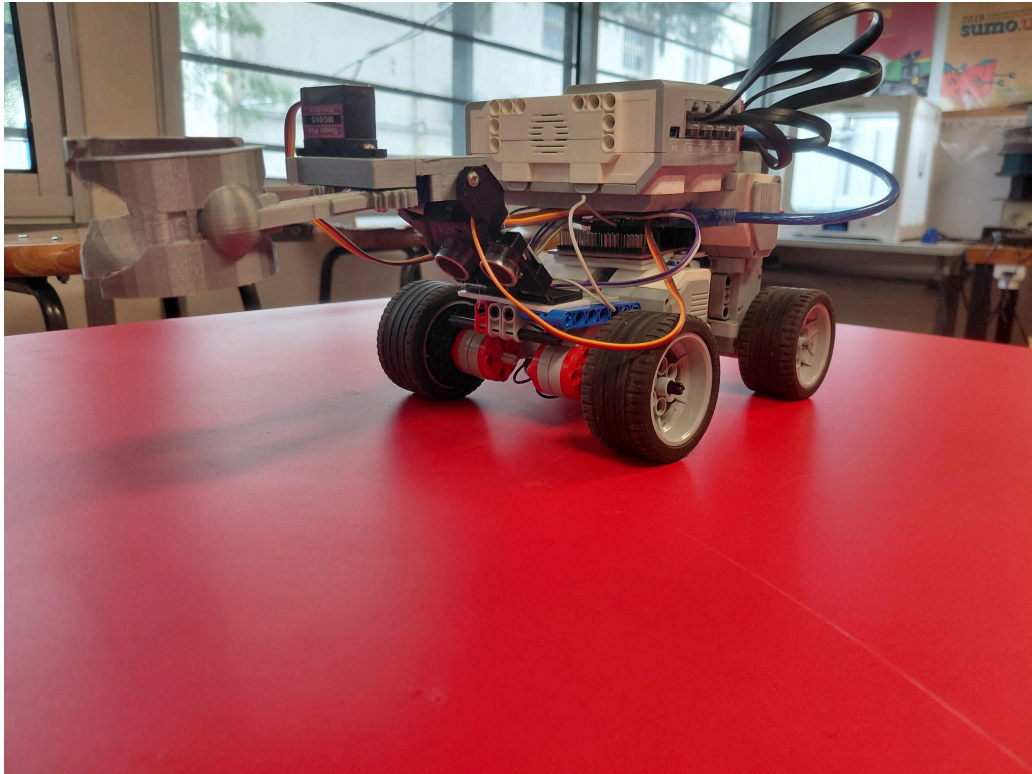
b.

Electronic Design and Manufacturing

- Provide a detailed overview of each aspect of the robot's electronic design and explain the used tools. Notably:
 - Sensors used
 - Main controller
 - Power subsystem
 - Actuators, etc
- Provide drawings, diagrams, or images to support your explanations and reasoning for your design choices. Reference requirements outlined during the project planning.
- Explain the testing procedures used to validate the design. Include relevant testing data.
- Highlight innovative and unusual solutions/approaches. Link back to constraints and requirements.

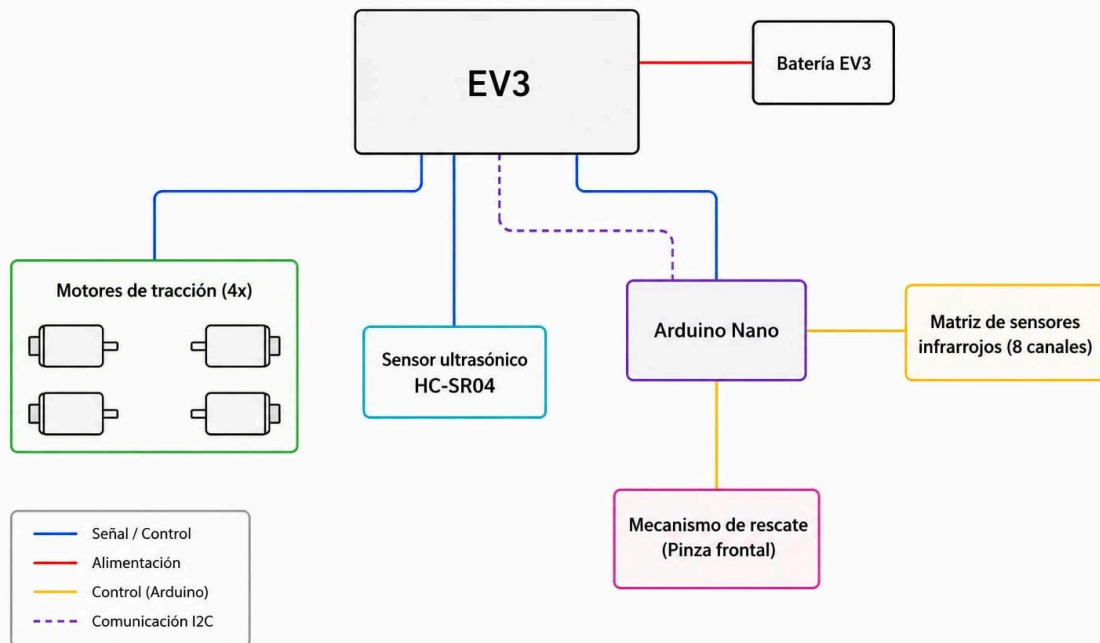
- **EV3 Infrared Sensor:** Responsible for detecting obstacles.
- **EV3 Color Sensor:** Responsible for recognizing colors through reflection (green, red, silver, black, and white).
- **EV3 Motors:** Responsible for the robot's movement along the track.
- **EV3 Brick:** Runs the program that allows the robot to process information and respond to different stimuli.
- **EV3 Parts:** Form the robot's structure by combining with the components mentioned above.
- **EV3 Cables:** Responsible for transmitting information between components.
- **USB Serial Connection:** Allows reliable communication between the EV3 and the Arduino.
- **EV3 Wheels:** Work together with the motors to move the robot.
- **Arduino Nuno:** Sends and receives information from the different components.
- **Servo Motors:** Help collect and distribute the victims.

- **Arduino Cables:** Distribute information to the connected components.
- **Sensor Array:** A set of sensors that measure reflected light.



The system consists of a single power source. The EV3 Brick powers the Arduino board, which requires a specific voltage and current to operate all of the robot's components correctly.

3.b Diagrama del Hardware



4-Software

- Provide an overview of the software.
- Do not include the source code in this document

The robot's software was developed using a combination of LEGO EV3 and Arduino Nano programming. The EV3 acts as the main control unit, handling navigation, line following, and the execution of different strategies during the competition. The Arduino Nano processes the information obtained from the infrared sensor array and sends it to the EV3 to facilitate faster and more accurate decision-making.

The line following algorithm uses data from the eight infrared sensors to determine the line's relative position to the robot. Based on this information, the system makes constant adjustments to the motor speeds to maintain a stable trajectory.

In addition, the software incorporates routines for obstacle detection using the ultrasonic sensor and for controlling the rescue mechanism. Communication between the EV3 and the Arduino allows both systems to work in a coordinated manner, exchanging information in real time throughout the course.

The software design focused on simplicity, reliability, and ease of adjustment, allowing for rapid modifications during the testing and optimization phases.

General software architecture

- Describe the general structure of your software
- Use diagrams and flowcharts to illustrate your explanations
 - Explain how the software/code solves any problems that arose during integration
- Describe the tools that your team used or developed (e.g. libraries, algorithms, AI models)

b.

The robot's software was organized in a modular fashion to facilitate its development, maintenance, and adjustment during testing. The system is divided into two main parts: the program running on the LEGO EV3 and the program running on the Arduino Nano.

The Arduino Nano is responsible for reading and processing the infrared sensor array. Once the line position information is obtained, this data is sent to the EV3 via communication between the two devices.

The EV3, in turn, receives the sensor information and executes the robot's main functions. These include line following, control of the drive motors, obstacle detection using the ultrasonic sensor, and activation of the rescue mechanism when necessary.

During the integration of the robot's various components, several challenges arose related to communication between the LEGO EV3 and the Arduino Nano, as well as the coordination between sensors, motors, and the rescue mechanism. To solve these problems, a modular software structure was developed that allows each component to

perform a specific task while exchanging information with the rest of the system. One of the main challenges was ensuring that the data obtained by the infrared sensor array was processed and sent to the EV3 quickly and reliably. To solve this problem, a communication system was implemented that allows for continuous updating of sensor data and its use for real-time navigation.

It was also necessary to coordinate the operation of the motors and sensors to avoid delayed responses or unstable movements. The software was fine-tuned through successive tests until it achieved smooth and consistent behavior during line following. Thanks to this approach, all the robot's components were able to work in an integrated manner, improving the accuracy, stability, and overall reliability of the system.

During the robot's development, LEGO Mindstorms EV3 and the Arduino IDE were used as the main programming tools. The system uses a line-following algorithm based on continuous readings from the infrared sensor array, allowing for constant corrections to the robot's trajectory. Routines were also developed for communication between the EV3 and the Arduino Nano, as well as for controlling the sensors, motors, and rescue mechanism. In addition, 3D design and printing tools will be used to manufacture custom robot components.

b.

One of the most innovative solutions implemented in our robot was the integration of LEGO Mindstorms EV3 and Arduino Nano. This combination allowed us to leverage the strengths of both platforms, distributing processing tasks and improving the overall efficiency of the system. While the Arduino processes the information from the infrared sensor array, the EV3 uses this data to control navigation and decision-making during the course.

To validate the software's functionality, numerous tests were conducted on training tracks that simulated competition conditions.

These tests included courses with sharp turns, abrupt changes in direction, intersections, and different line configurations. During each trial, the robot's behavior was observed, and adjustments were made to the line-following algorithms and the communication between the EV3 and the Arduino.

The results obtained show a progressive improvement in the robot's stability and accuracy. As the tests progressed, tracking errors decreased, and a faster response to changes in the course was achieved. Furthermore, the testing sessions allowed us to detect integration problems and optimize various software parameters, contributing to a more robust and reliable system to face the challenges of competition.

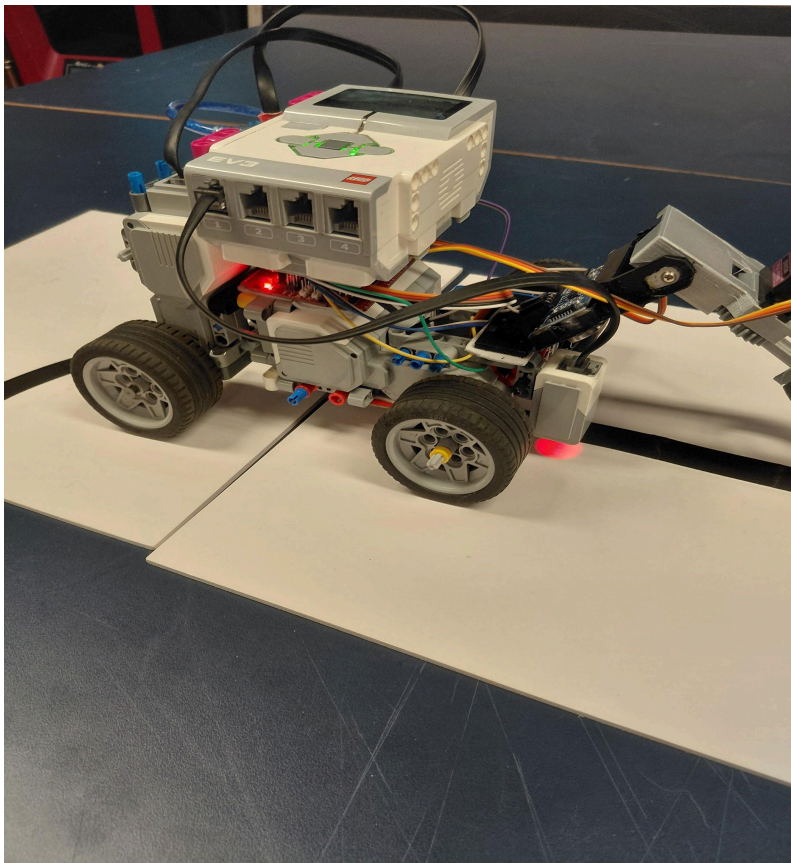
5-Performance evaluation

- Evaluate the performance of the robot against competition challenges
- Describe the testing procedures implemented to verify the robot's performance
- Explain how the test results were analyzed and how they impacted the development

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6-Conclusion

- Brief conclusion of this paper.

The whole team put in a lot of effort this time and we met most of our goals, such as having a more organized team, we achieved greater consistency in our work. We worked hard on this project, there were things to improve, but all the challenges we overcame were successfully resolved.

Appendix (optional)

- N.B.: The appendix will NOT be evaluated towards the score. The appendix should only be used to provide external links.

References

- References to external sources used for major parts of the development process.